Robust Human Detection Within a Highly Dynamic Aquatic Environment in Real Time

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***Abstract*—This paper presents a real-time foreground detection method for monitoring swimming activities at an outdoor swimming pool. Robust performance and high accuracy of detecting objects-of-interest are two central issues of concern. Therefore, in this paper, a considerable amount of attention has been placed on the following aspects: 1) to establish a better method of modeling aquatic background, which exhibitis dynamic characteristics with random spatial movements, and 2) to establish a method of enhancing the visibility of the foreground by removing specular reflection at nighttime. First, the development of a new background modeling method is reported. In the proposed approach, the background is modeled as a composition of homogeneous blob movements. With an implementation of a spatial searching process, the proposed method shows capability in associating and distinguishing movements caused by the background. Hence, this contributes to better performance in foreground detection. On the issue of enhancing the visibility of the foreground, a decision-based filtering scheme is proposed as a preprocessing step. A defined concept term, fluctuation measure, is defined for classifying each pixel to be one of the predefined types. This has allowed suitable spatial or spatiotemporal filters to be applied accordingly for color the compensation step. All of these developments are evaluated by testing live on a busy Olympic-size outdoor public swimming pool. Both qualitative and quantitative evaluations are reported. This provides a comprehensive study of the system.**

***Index Terms*—Adaptive background estimation, background and foreground modeling, dynamic aquatic background, real-time video surveillance, robust human detection, spatiotemporal filtering, thresholding with hysteresis.**

I. Introduction

AUTOMATED video surveillance [1]–[4] addresses the challenge to perform real-time analysis and constant monitoring of activity. The intelligence to detect abnormal events and the capability to perform monitoring tasks in semi/full automation has brought along advantages in the aspects of improving safety in our surroundings and the quality of work of security personnel. Along with the advances in algorithm and technique development, surveillance technology has found applications in various sectors. For instance, remote surveillance of unattended environments has received growing attention at places like airports [5]–[7], highways [8], [9], railway infrastructures [10], parking lots, and on the road [11]–[14]. As reported, varied degrees of success have been achieved.

Besides applications on the ground, as mostly considered in current literature, the problem of surveillance in aquatic environments is found to be equally important. One potential application is to provide constant monitoring of swimmers' activities at swimming pools. Such a system can serve as additional "eyes" in electronic modes to assist the monitoring process, on top of the watchful eyes of lifeguards. With such motivation, the objective of this paper is to develop a framework to achieve robust swimmer (foreground) detection under long-hour operation in the environment of a public swimming pool. This work addresses the fundamental steps before it is able to proceed to the real-time recognition of high-level swimming activities.

The problem with building a vision-based surveillance system for pool environments is found to be relatively new with limited literature. Related important works could be referred to a few commercially patented systems reported in [15]–[18], which attempted to address the problem by using underwater cameras. In these systems, an alert is triggered if a drowning victim is found motionless at the bottom of the pool over an extended period of time. Despite numerous capabilities by these systems, the reliance on underwater cameras has, in the meantime, brought along a few concerns. One of the concerns is that a slow drowning detection response is expected. This is due to the detection which is confined to the victim found at bottom of pool, which is at a later stage of drowning. Besides, higher installation and maintenance costs will be expected, as a specially fabricated apparatus is required for underwater environments. In addition, there is a limitation in deploying these systems in shallow pools because of possible blocking of camera views by standing swimmers.

Motivated by promising results obtained based on our preliminary works [19]–[22], in this paper, we explore the development of new methods for processing videos captured using overhead cameras, in contrast to using underwater camera. Monitoring a swimming pool from its top view brings the advantage that it is always possible to capture and analyze swimmers' activities at the water's surface. Therefore, this allows the detection of any early drowning symptoms when the victim is struggling at the water's surface. When considering such a live system, we face compounded challenges unique to aquatic environments. For instance, there are continual movements of back-

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ground elements due to disturbances at the water's surface (e.g., movements of floor patterns), which could be easily misidentified as foreground. Additionally, long-hour operation has also exposed the problem to a wide range of lighting variations. For instance, artificial lighting at nighttime casts camouflage effect on the foreground. Therefore, in this paper, we first carry out studies on: 1) the unique characteristics of aquatic environments and 2) technical challenges faced to segment foreground objects from such dynamic aquatic backgrounds. A novel foreground detection method, comprising a set of algorithms, for long-hour operation is developed based on such studies.

The first contribution of this work is the development of a new background modeling method. Since the aquatic background exhibits random homogeneous blob movements, we suggest the modeling of the background as a composition of homogeneous region units. With the implementation of a spatial searching mechanism, long-range spatial dependency of the background pixel is efficiently addressed. This develops the capability to distinguish between changes caused by foreground movements and background movements. Moving one step further, a scheme that combines thresholding with hysteresis [23], denoising, and connected component grouping [24] is devised. This addresses the problem of selecting appropriate thresholds for maximizing foreground detection while minimizing foreground noises. In addition, to maintain a robust performance at nighttime, a preprocessing filtering method is developed for handling reflections caused by strong artificial lighting. Through classifying pixels of the current frame into different identified pixel types, spatiotemporal filters are applied accordingly for frame enhancement.

This paper is organized as follows. Section II presents a detailed review of relevant works that places our work in perspective, and highlights numerous technical challenges that differentiates our problem from problems discussed previously. Section III details the proposed region-based background modeling and thresholding operation that combines thresholding-with-hysteresis principles, denoising, and connected component grouping. Section IV describes the proposed nighttime processing module that comprises a decision-making module and a filtering module for recovering swimmer's pixels. Last, in Sections V and VI, experimental results and some concluding remarks are presented, respectively.

II. Our Work in Perspective

In this section, we review background subtraction algorithms that are relevant to our method and discuss various unique issues faced to put our efforts in perspective. Fairly extensive reviews on other existing background subtraction algorithms could be referred to the review papers by Boult *et al.* [25], Elgammal*et al.* [26], and Wang *et al.* [27].

A. Previous Work on Background Subtraction

The simplicity and robustness applying to a wide range of situations are advantages of the *background subtraction* approach, making it a popular technique adopted among the literature of automated video surveillance. Its basic steps involve: subtracting a current frame from a constructed background representation followed by thresholding the derived difference image to recover foreground pixels which usually correspond to larger pixelwise differences. To yield good performance, a good modeling of background and foreground of a monitored scene is highly essential. In the following, we broadly classify the existing background subtraction algorithms into two categories: 1) pixel-based methods and 2) block-based methods, and we discuss the pros and cons of each. Such classification is to provide a clear picture about the contribution of this paper.

Pixel-based methods could be loosely referred to as the approach of modeling background being a composition of independent pixel-based processes. One example of such methods is the Pfinder system [28]. This work demonstrated successful real-time tracking of one single person within a controlled indoor environment by constructing each background pixel based on a single Gaussian distribution in YUV color space. The method recursively updated each background pixel model over time to adapt to possibly slow illumination changes caused by variations in ambient lighting. In [29], the same principle has been adopted, but a Kalman filter was used instead for background updating process.

When addressing outdoor surveillance which experiences more dynamic background movements, it is often observed that there are elements of different background objects being projected onto the same pixel. For instance, one pixel could be the image of the sky at one frame but a tree leaf in the following, which exhibits totally different statistical properties. To address a higher dynamic property of outdoor environments, recent works of pixel-based approaches by Stauffer*et al.* [11], [12] and Friedman *et al.* [13] proposed the modeling of backgrounds using mixture of Gaussian (MoG) distributions. In [11] and [12], pixel intensity was modeled as a mixture of k-Gaussian distributions. In [13], a mixture of three Gaussian distributions was defined in advance to model three possible components of the monitored scene, corresponding to road, shadow, and vehicle. Nonparametric modeling is also popular in [14], [25], and [26]. For example, the work of W4 [14] modeled the background by representing each background pixel process by its minimum and maximum intensity, as well as the maximum intensity difference of two consecutive frames. Meanwhile, the work by Elgammal *et al.* [26] suggested an alternative way by utilizing the principle of kernel density estimation for background modeling.

One common limitation of pixel-based techniques described above is that they are not meant for the purpose of addressing the situation with significant background movements. Thus, these methods are generally more confined to environments with less background movement, e.g., within controlled indoor and outdoor environments.

Apart from pixel-based methods, the works by Hsu *et al.* [30] and Matsuyama *et al.* [31] suggested the use of the statistical features of divided image blocks for change detection. The classic paper by Hsu *et al.* [30] introduced a method of detecting significant changes in image blocks by fitting a second-order bivariate polynomial. Meanwhile, in [31], statistical measures were defined to quantify the correlation among blocks. The need

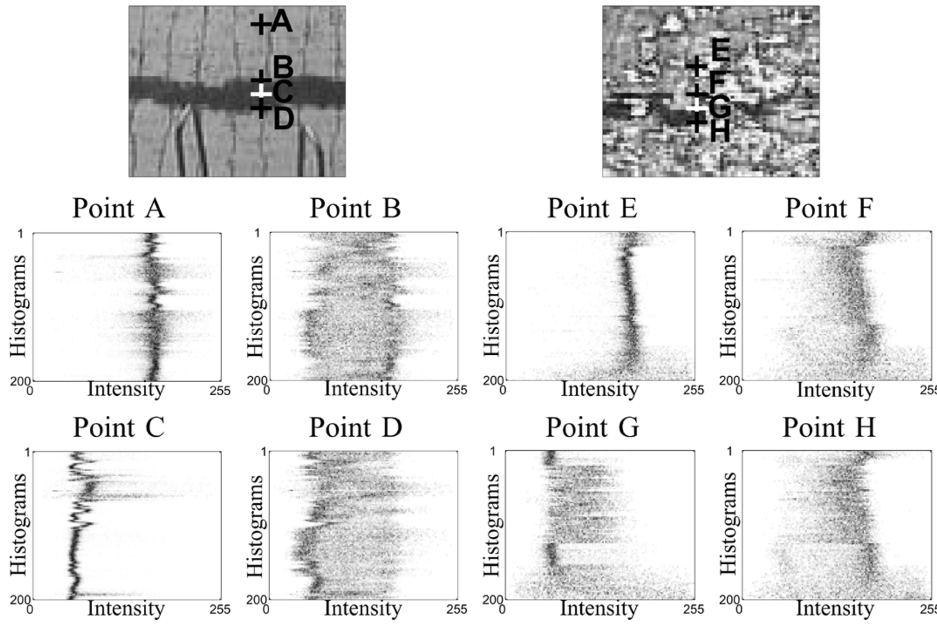


Fig. 1. Different portions of a typical pool: intensity histograms of pixels over time. Wide distributed histograms are due to rapid movements of pixels belonging to background elements.

of using image blocks is one limitation of the block-based technique that has constrained the detection to be in a block-based unit. As a result, this method is limited to coarse detection.

It is also worth noting that there are fundamental works looking into overcoming distortion caused by the the water surface based on physical considerations [32]–[34]. In [32], the surface shape of a nonrigid transparent object (such as water) was reconstructed from the apparent motion of observed pattern. In [33], a model-based approach was proposed to recover the shapes and the poses of transparent objects from known motion. In [34], an approach that recovered surface shapes of transparent objects was developed based on polarizing filter. Definitely, these works provide valuable insights, where the effort of image reconstruction could be viewed as a preprocessing step for better foreground detection. However, as most of these approaches are rather computation intensive, which are not meant for real-time surveillance application, our focus of this work is to extend the current surveillance technique to handle more dynamic backgrounds, while taking into account some physical considerations as highlighted in [32]–[34].

Our proposed method described here differs from the current background subtraction methods mentioned previously, where we model the background using a set of block-derived homogeneous region-based units, as opposed to pixel-based or block-based approaches.

B. Challenges Faced

A study of the statistical property of the problem is illustrated in Fig. 1. It shows an analysis of the intensity variation of pixels at various positions at the scene over time. The horizontal axis of the figure represents the gray level intensity of pixels while each row in the vertical axis represents the intensity histogram of each pixel over 200 frames. Each subsequent row depicts the histogram of that pixel over the next 200 frames; hence, the plot captures the pixel's intensity patterns over a total of 40 000 frames. The wide distribution of the intensity histogram demonstrates a higher level of complexity and challenge involved in our problem compared with the problem of detecting humans on the ground, which usually has a much less distributed intensity, as shown in [25, Fig. 6].

Apart from the above issues unique to the aquatic environment, there are also common problems faced in outdoor surveillance, such as continual illumination changes due to ambient lighting and auto gain effects of cameras. A fast background updating scheme is important to adapt to such illumination changes. This, however, will also exacerbate the problem of foreground objects blending into background models due to segmentation errors. The corruption of background models will, in turn, lead to more segmentation errors on the subsequent frames.

Other technical challenges include the need for an algorithm that runs in near real-time detection and one that can be implemented using low-cost and off-the-shelf hardware platforms.

C. Our Contributions

The unique features of this work that distinguish it from other present works are as follows.

- An outdoor aquatic surveillance problem, which involves automated human segmentation within hostile aquatic environments, is considered. New insights into typical technical challenges faced in considering outdoor surveillance are discussed. In particular, problems unique to human detection within dynamic aquatic environments are detailed.
- The developed human segmentation method for aquatic environments provides the foundation of building an automated aquatic surveillance system using overhead cameras.
- To address the highly dynamic property of aquatic background, we have proposed a homogeneous region-based

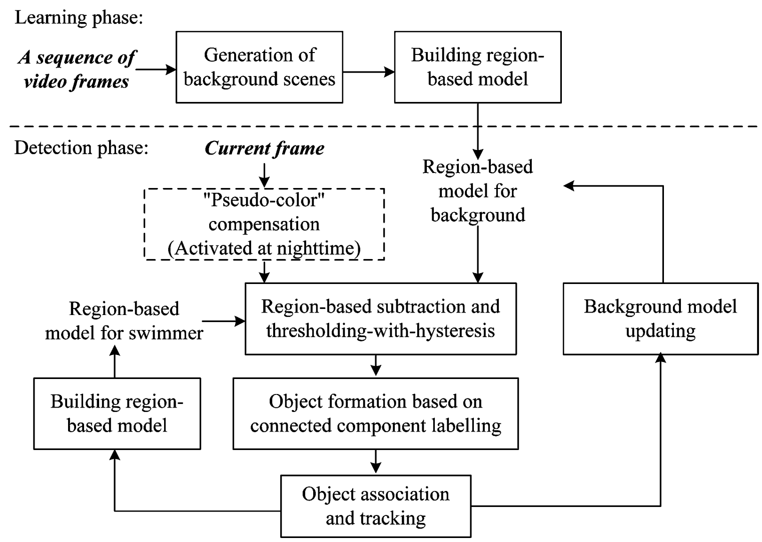


Fig. 2. Architecture of the proposed foreground detection algorithm.

modeling scheme that models the background as a composition of homogeneous region processes. This facilitates the implementation of an efficient spatial searching scheme that exploits long-range spatial displacement of background pixels to capture background movements.

- A preprocessing scheme has also been proposed that enhances the detection of swimmers who are partially hidden by specular reflection at nighttime. This has been achieved with a defined fluctuation measure concept for pixel classification, and the appropriate use of spatiotemporal filters for frame enhancement.
- Our current approach also shows significant improvements made over our previous works reported in [21] and [22]. In our previous works, models for various possible background components (i.e., water, lane dividers, and swimmers) within the swimming pool are explicitly modeled. Such an approach has failed to operate continuously for long periods of time in the live system. This is due to complexity involved in building sufficient representative models to accurately describe the background and foreground.

III. Proposed Methodology

The proposed methodology for foreground detection comprises: 1) a learning module, 2) a detection module, and 3) a spatiotemporal filtering module, with the architecture as presented in Fig. 2. The following two sections, respectively, describe the proposed learning module and detection module. The proposed spatiotemporal filtering scheme will be detailed in Section IV.

A. Region-Based Background Subtraction Scheme

The basic concept of the proposed region-based method is to model background as a composition of homogeneous region processes. The modeling is to account for nonstationary property of each region that experiences random positional variations, as well as illumination and color variations.

Given a sequence of $N_1 \times N_2$ background frames, homogeneous background regions are formed by taking the steps: Each of these background frames is divided into $n_1 \times n_2$ nonoverlapping square blocks with size of $s \times s$ each, where $n_1 = (N_1/s)$ and $n_2 = (N_2/s)$. Clustering operation using hierarchical k-means [24] is performed on color vectors collected from these divided square blocks to form homogeneous regions. Let $\boldsymbol{X}^s_{a,b}$ be a set of color vectors collected from square blocks (a,b) of all these background frames, where $1 \le a \le n_1$ and $1 \le b \le n_2$. Homogeneous background regions are generated by clustering every $\boldsymbol{X}^s_{a,b}, \forall a,b$ that will group each $\boldsymbol{X}^s_{a,b}$ into homogeneous regions $\{\boldsymbol{R}^1_{a,b},\ldots,\boldsymbol{R}^c_{a,b}\}$. Statistical representation of background is established based on constructed statistical models of all the homogeneous regions formed.

Fig. 3 illustrates a study of luminance histogram of pixels collected from different square blocks as marked at the images on the left. Multiple peaks are observed when there is more than one background element within the same block. For example, two distinct peaks are observed for the cases of blocks B and D that contain both elements of water and lane divider. Composition of a few single multivariate Gaussians (in the three-dimensional color space) forms a MoG-like distribution for each block as shown. The clustering step produces the effect of breaking down complicated signals of a block into homogeneous region processes, where each could be reasonably modeled using one single multivariate Gaussian distribution.

Mathematical Representation: Denote the mean and covariance matrix of a homogeneous region $\boldsymbol{R}^k_{a,b}$ by $\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}} = \{\mu^1_{\boldsymbol{R}^k_{a,b}},\ldots,\mu^d_{\boldsymbol{R}^k_{a,b}}\}$ and $\Sigma_{\boldsymbol{R}^k_{a,b}}$, respectively, where $d = 3$ is the dimension of the color space. The probability that a pixel $\boldsymbol{x}_{i,j} = \{x^1_{i,j},\ldots,x^d_{i,j}\}$ belongs to a homogeneous background region $\boldsymbol{R}^k_{a,b}$ could be described mathematically by

$$P(\boldsymbol{x}_{i,j}|\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}},\Sigma_{\boldsymbol{R}^k_{a,b}}) = \frac{1}{(2\pi)^{d/2}|\Sigma_{\boldsymbol{R}^k_{a,b}}|^{1/2}} \times \exp\left\{-\frac{1}{2}(\boldsymbol{x}_{i,j}-\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}})\Sigma^{-1}_{\boldsymbol{R}^k_{a,b}}(\boldsymbol{x}_{i,j}-\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}})^T\right\}. \quad (1)$$

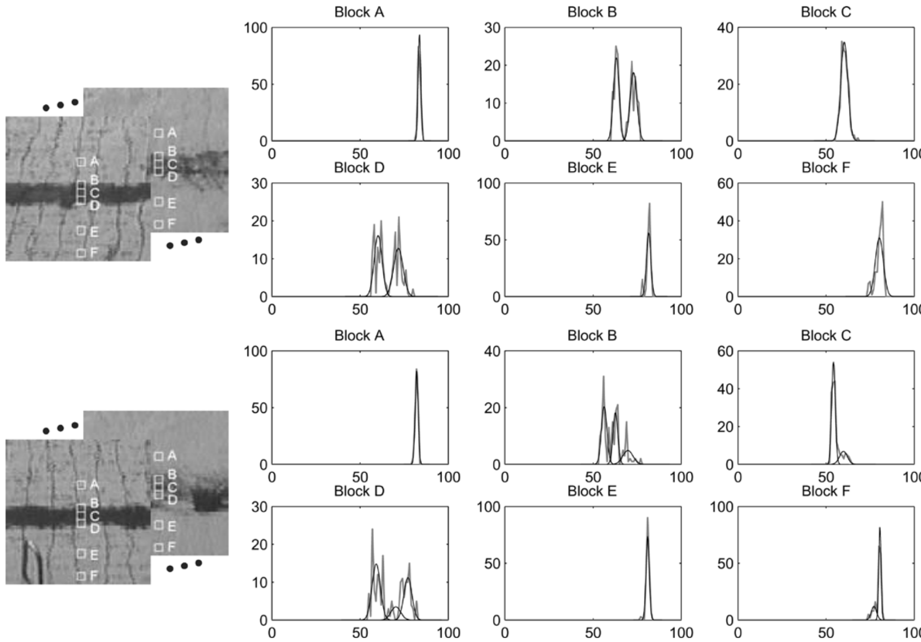


Fig. 3. Luminance histograms of pixel intensities extracted from Blocks A–F as marked at two typical example images on the left, showing multiple peaks when more than one object is projected onto the block over time. Modeling each homogeneous region as one single-Gaussian distribution demonstrates a good approximation of the original signal.

Without compromising much the accuracy at final segmentation, we assume that pixel values in different dimensions are independent to reduce computational complexities. Based on such an assumption, we further simplify the probability measure in (1) to be

$$P(\boldsymbol{x}_{i,j}|\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}, \boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}) = \prod_{m=1}^{d} \frac{1}{\sqrt{2\pi}\sigma^m_{\boldsymbol{R}^k_{a,b}}} \exp\left\{-\frac{(x^m_{i,j} - \mu^m_{\boldsymbol{R}^k_{a,b}})^2}{2(\sigma^m_{\boldsymbol{R}^k_{a,b}})^2}\right\} \quad (2)$$

where $\boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}} = \{\sigma^1_{\boldsymbol{R}^k_{a,b}}, \ldots, \sigma^d_{\boldsymbol{R}^k_{a,b}}\}$ is the standard deviation matrix of $\boldsymbol{R}^k_{a,b}$. Such n assumption is reasonable, as we have chosen the CIELa*b* color space, where the luminance component is well decorrelated from chronimance components. Thus, the similarity of a pixel with respect to a homogeneous region of the background could be quantified mathematically using the distance measure as follows:

$$D(\boldsymbol{x}_{i,j}|\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}, \boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}) = \sqrt{\frac{\sum_{m=1}^{d} (x^m_{i,j} - \mu^m_{\boldsymbol{R}^k_{a,b}})^2}{(\sigma^m_{\boldsymbol{R}^k_{a,b}})^2}}. \quad (3)$$

The normalization by the denominator in (3) provides an adaptation to any change of background regions over time. This is facilitated through updating step of $\boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}$.

As noted, background frames are required to establish background models. However, in practice, it may not be possible to collect a sequence of empty background scenes, as most scenes captured probably contain both foreground and background. The following sections describe in detail the implementation in the aspects of: 1) generating a sequence of background frames during the learning phase, 2) formation of homogeneous region-based background models, and 3) recursively updating the established background models to adapt to environmental changes.

a) Generating background frames: Generating a sequence of "clean" background frames is the first step of the background learning process. In this section, we propose the formation of background frames based on a temporal vector median filter and a coarse swimmer skin model.

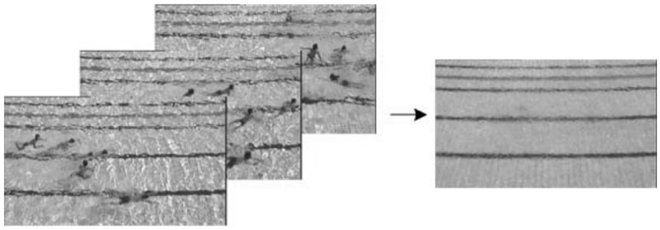
When analyzing a collected array of pixels' values at a position in the frame over time, the presence of swimmer pixels could be regarded as impulsive noise among other collected background pixels. Therefore, it is removable by applying a vector median filter. However, the filtering step is found insufficient if there are swimmers that are stationary during the filtering period. Thus, we first apply a skin color model as a preprocessing step to isolate swimmer pixels from the background formation process. Then, residual foreground pixels are removed using a temporal vector median filter, which involves the following steps.

Let $\boldsymbol{X}^\tau_{i,j}$ be an array of color vectors collected over τ number of frames, i.e., $\boldsymbol{X}^\tau_{i,j} = \{\boldsymbol{x}^t_{i,j}|t = 1, \ldots, \tau\}$, where $\boldsymbol{x}^t_{i,j}$ is the color vector of the tth image at position (i, j). The sampling rate that determines the temporal interval between two $\boldsymbol{x}^t_{i,j}$ was decided empirically after considering a tradeoff between duration needed for the learning phase and efficiency to remove swimmer pixels. A background frame is generated by performing vector median filtering on $\boldsymbol{X}^\tau_{i,j}, \forall\, i, j$ that generates

$$\boldsymbol{B}^1 = \{\boldsymbol{y}_{i,j}|i = 1, \ldots, N_1 \text{ and } j = 1, \ldots, N_2\} \quad (4)$$

where $\boldsymbol{y}_{i,j}$ is obtained based on

$$\boldsymbol{y}_{i,j} = \left\{\boldsymbol{x}^q_{i,j} \in \boldsymbol{X}^\tau_{i,j} \left| \min_{\boldsymbol{x}^q_{i,j}} \sum_{r=1}^{\tau} \left| \boldsymbol{x}^r_{i,j} - \boldsymbol{x}^q_{i,j} \right| \right.\right\}. \quad (5)$$



(a) (b)

Fig. 4. Generation of a background scene using a temporal vector median filter. (a) A sequence of frames contains foreground swimmers. (b) Generated background scene.

As evident from Fig. 4, the above operation has efficiently generated a "clean" background frame in an automated manner. The subsequent background frames $\{\boldsymbol{B}^u | u = 2, \ldots, \kappa\}$ are generated by maintaining an overlapping sliding window that captures pixel vector $\{\boldsymbol{x}_{i,j}^t | t = u, \ldots, \tau + u\}$, $\forall i, j$. In the process, κ number of background frames are generated.

b) Generating initial background models: From our experiments, we found CIELa*b* color space to produce better swimmer segmentation results compared with other color spaces. Thus, before the background formation steps, generated background scenes are first converted into the CIELa*b* space, forming matrix $\{\boldsymbol{B}_{\text{Lab}}^u | u = 1, \ldots, \kappa\}$. Two steps are taken to establish the proposed region-based background representation. 1) Divide each $\boldsymbol{B}_{\text{Lab}}^u$ into $n_1 \times n_2$ of nonoverlapping $s \times s$ square blocks. 2) Apply a hierarchical k-means **[24]** on each $\boldsymbol{X}_{a,b}^s$ to obtain cluster centers $\boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}$ and standard deviations $\boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}$ of homogeneous regions $\boldsymbol{R}_{a,b}^k$.

The clustering process starts by initiating one dominant data cluster. In the subsequent iterations, smaller and more compact data clusters are formed through splitting until the distance of any two closest cluster centers is smaller than a threshold or the number of clusters reaches a predetermined value, whichever is achieved first. Therefore, this forms c-number of homogeneous regions $\boldsymbol{R}_{a,b}^k$ and yields the initial background models $\boldsymbol{C}_{a,b}$ to be

$$\boldsymbol{C}_{a,b} = \begin{bmatrix} \boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k} \\ \boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k} \end{bmatrix}, \text{ where } k = 1, \ldots, c. \tag{6}$$

Since we have assumed each color channel is independent, parameters $\boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}$ and $\boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}$ are obtained by computing the means and standard deviations of color pixels in the respective color channel.

c) Updating background models: Recursively updating the established initial background models is essential in order to adapt to changes in outdoor environment. For example, there are illumination changes due to sudden blocking of sunlight by clouds or significant ripple movements on the water's surface due to wind. In this section, we suggest a recursive linear interpolation scheme for continuously updating the background models, which shares some similarities as the scheme used in [11], [12]. After collecting a number of pixels belonging to $\boldsymbol{R}_{a,b}^k$, parameters $\boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}$ and $\boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}$ at time-t is updated based on

$$\boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}^t \leftarrow (1-\rho)\boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}^{t-1} + \rho \boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}^t \tag{7}$$

$$\boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}^t \leftarrow (1-\rho)\boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}^{t-1} + \rho \boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}^t \tag{8}$$

where $\rho = 1/\kappa$ is the learning factor for adapting current changes, and κ has been defined previously in Section III-A, which is the number of "clean" background frames generated for generating initial background models. Equivalently, we could interpret (7) and (8) as the averages of all $\boldsymbol{\mu}_{\boldsymbol{R}_{a,b}^k}^t$ and $\boldsymbol{\sigma}_{\boldsymbol{R}_{a,b}^k}^t$ of $\boldsymbol{R}_{a,b}^k$ over a sliding window of length κ. The updating process is performed by only considering pixels within a threshold distance from the mean. This is to ensure the Gaussian model is being adopted slowly, while reducing any chance to include outlier that would corrupt the statistical property.

In addition, the ability to create and destroy unnecessary distribution is also essentially required to maintain a dynamic background modeling. Each distribution is given a κ number of lifespan. If there is no pixel being classified to any existing distribution, it would be eliminated after κ frames. New gaussian distribution is created based on those unmatched pixels. To avoid any corruption by noise, the new distribution is created only if its color standard deviations are smaller than a typical value, which is determined experimentally.

B. Foreground Swimmer Detection and Modeling

One main problem of foreground detection is to attain a good tradeoff between maximizing target detection while minimizing false positive. In this section, we describe a scheme that balances both requirements by incorporating active background movements and an enhanced thresholding scheme.

Let $\boldsymbol{x}_{i,j}$ be color vector of pixel at location-(i, j) in the square block-(a, b). Using (3) and considering possibly movements of pixel within a local window, we define the color discrepancy between a pixel and its neighboring background models to be

$$D_{i,j}^{\min} = \min\{D(\boldsymbol{x}_{i,j} | \boldsymbol{\mu}_{\boldsymbol{R}_{a+q,b+r}^k}, \boldsymbol{\sigma}_{\boldsymbol{R}_{a+q,b+r}^k})\} \tag{9}$$

where $q, r = -v, \ldots, v$ and the dimension of the local window is $(2v+1) \times (2v+1)$ in square-block unit. Thresholding on $D_{i,j}^{\min}$ $\forall$ i, j uses the following operation:

$$M_{i,j} = \begin{cases} 0 \text{ background}, & D_{i,j}^{\min} < \alpha \\ 1 \text{ foreground}, & \text{otherwise}. \end{cases} \tag{10}$$

This yields a binary segmentation map, $\{M_{i,j} | i = 1, \ldots, N_1 \text{ and } j = 1, \ldots, N_2\}$, which contains "0"s and "1"s marking background and foreground, respectively.

Figs. 5 and 6 provide studies of various parameter settings of local window size by variable v and thresholding variable α. Fig. 5 shows $\{M_{i,j}\}$ when varying α by fixing v. Looking from left to right, consistent detection performance is observed across various situations of the background. This demonstrates the robustness of the developed algorithm to adapt to different situations. Meanwhile, when looking down the rows, the increase in α value corresponds to fewer foreground noises, but with a less accurate foreground boundary segmentation. Therefore, it is clear from the figure that a single thresholding mechanism is inherently less efficient.

It is also noted from Fig. 6 that the consideration of neighboring blocks when computing $D_{i,j}^{\min}$ has demonstrated significant reduction in mis-classification errors at regions, such as, areas around lane dividers and seemingly moving shadows. The

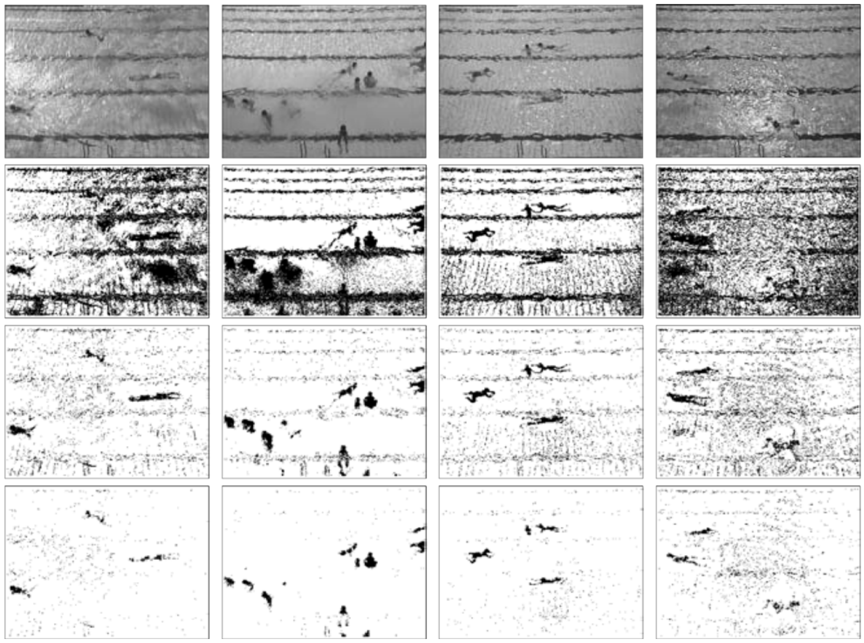


Fig. 5. Comparison of foreground detection with α ranges from small value to large value, looking down the rows. Consistently good detection performance is achieved when applying on various situations, looking from left to right.

Original image

$v = 0$

$v = 1$

$v = 2$

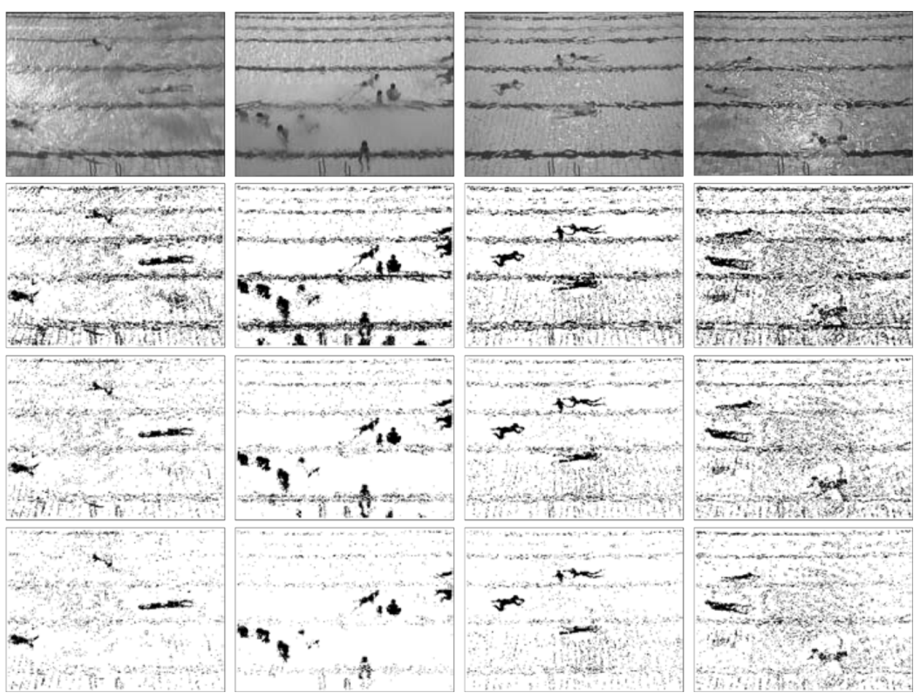


Fig. 6. Comparison of foreground detection for cases: (from the second to fourth rows) without any neighborhood structure, with eight nearest-neighborhood structure, and with 24 nearest-neighborhood structure. Increasing the search space gives better performance in capturing background movements as observed across various situations.

increase in search space effectively incorporates background movements across various situations; hence, reducing false positives.

1) Choice of Threshold, α: In this section, a solution that combines the principles of thresholding with hysteresis [23], denoising, and connected component grouping [24] is developed as a better alternative to the intuitive single threshold scheme.

The thresholding step is performed in a hierarchical manner, as illustrated in Fig. 7. A similar framework was also proposed in [25]. In our proposed framework, a low-threshold map is first generated using (10) based on a low threshold value α_l. As evident from Fig. 7(a), the low-threshold map yields a good segmentation of the foreground objects. However, it also comprises lots of foreground noises. Based on the same operation, a high-threshold map is generated with less foreground noise by using a higher threshold value α_h. To further suppress foreground noise, a coarser resolution of the high-threshold map, termed the *parent map* is obtained. Elements in the coarse res-

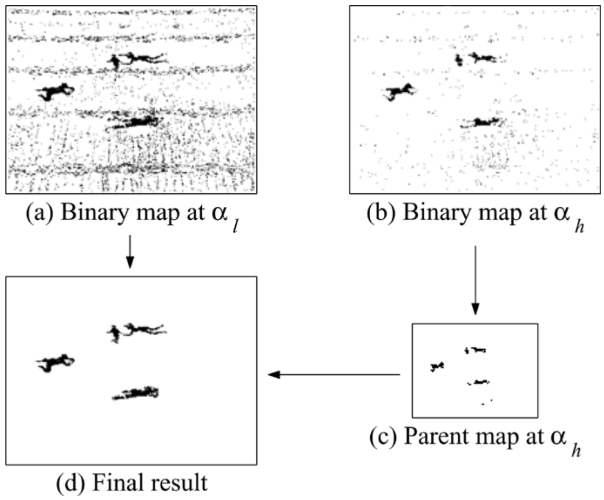


Fig. 7. Thresholding for foreground swimmer detection that combines principles of thresholding with hysteresis, denoising, and connected component grouping.

olution parent map are labeled "1" if there is a majority of corresponding binary pixels at its full resolution with $D_{i,j}^{\min} > \alpha_h$.

To form the final detection map, the low-threshold map is compared with the parent binary map. A foreground is formed if a connected region of pixels has $D_{i,j}^{\min}$ above α_l and its corresponding area in the parent map also form smaller connected blob(s). Such steps achieve an enhanced detection with a lower number of false positive as shown in the final result given in (d).

2) Updating Swimmer Models: In the same way as forming the background model, once a swimmer is detected, a region-based swimmer model is also defined and updated every frame. During the online processing phase with the establishment of foreground models, color discrepancy measure between a pixel's color and neighboring foreground models are computed using

$$D_{i,j}^{f\min} = \min\{D(\boldsymbol{x}_{i,j}|\boldsymbol{\mu}^f_{\boldsymbol{R}^k_{a+q,b+r}}, \boldsymbol{\sigma}^f_{\boldsymbol{R}^k_{a+q,b+r}})\} \tag{11}$$

where $\boldsymbol{\mu}^f_{\boldsymbol{R}^k_{a+q,b+r}}$ and $\boldsymbol{\sigma}^f_{\boldsymbol{R}^k_{a+q,b+r}}$ are mean and standard deviation matrices of foreground models. Thus, pixels with $D_{i,j}^{f\min} < D_{i,j}^{\min}$ are first classified as the highest confidence swimmer pixels. The remaining swimmers pixels are detected using the thresholding-with-hysteresis scheme detailed earlier. The construction of foreground model is particularly useful to enhance the performance of detecting foreground.

IV. Swimmer Detection With Filtering Scheme at Nighttime

Operation at night in aquatic environment faces additional challenges posed by glare and reflection from artificial lighting. In this section, we first describe the study of using a polarizer in our problem. Then, we introduce the suggested filtering method for reflection removal.

A. Use of Polarizer for Reflection Removal

The use of polarizer has been a well-known photographic technique to suppress virtual layer and reflection [35]. As discovered in several works, if the position and orientation of camera as well as the reflecting surface are known, the use of polarizer could be a simple yet efficient method in removing reflection by setting camera at the Brewster angle [36]. In [36], the work demonstrated the success of separating transmitted and reflected images off a semi-reflecting medium (e.g., a glass window). This was achieved through accurately estimating the inclination angle. In [37], an electrically controllable polarizing filter was developed, which could automatically reduce undesirable reflected light by estimating the polarization state of incident light. The method did away with the need to manually rotate the polarizer while claiming the capability of removing 80% of reflected light for some experiments.

A study of using polarizers in our problem has been performed as demonstrated in Fig. 8. The figure shows a comparison of different sample images captured with and without polarizers at different inclination angles of camera and polarization orientations. As evident from the figure, the use of a polarizer does not bring a perfect solution to our problem, where roughly equal amounts of reflection are still observed. Referring to the second row of the figure, there is still a significant amount of reflection observed when the camera is positioned roughly at the Brewster angle, which is approximately 40° looking down from the horizon.

We believe that polarizer could be useful in many aspects. However, we face difficulty achieving good results in our system. Due to the inaccessibility of camera, which is mounted high at building structure, we also face additional difficulty adjusting the polarizer. Therefore, we have resorted to a filtering approach for reflection removal.

B. Filtering Approach for Reflection Removal

We view the problem of removing reflection as detecting a camouflaged target, which could be tackled through exploiting spatiotemporal information. There have been numerous successful cases as found in the literature. For instance, spatial analysis was used in [25] to compensate missing pixels of partially camouflaged foreground objects from a static background. Also, temporal information is used to detect foreground objects under time-varying illumination [38].

The main idea of the filtering method described here is to first perform pixel classification. Subsequently, appropriate filtering actions could be applied accordingly for different class of pixels.

1) Pixel Classification Based on a Fluctuation Measure: Extensive statistical analysis in the temporal domain shows the observations that: 1) most background pixels will have low-pixel color fluctuation; 2) the main parts of a moving foreground, dynamic parts of the background and reflections, will have a medium level of pixel color fluctuation; and 3) fast-moving parts of the foreground and acquisition noise will both have high-pixel color fluctuation. This suggests that the rate of pixel color fluctuation in the temporal domain provides a useful property that could be exploited for the classification of pixels into different classes of having different characteristics.

As also noted, there is a close correlation between the pixel color fluctuation and the pixel intensity fluctuation; we have simplified the problem by working on pixel intensity for the

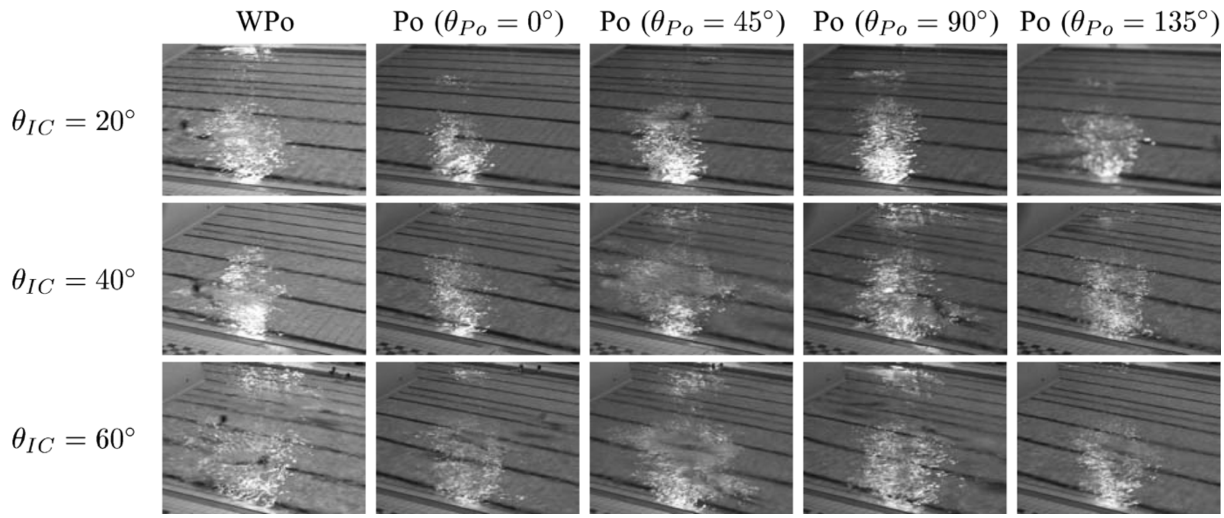


Fig. 8. Sample images captured with and without polarizer at different camera inclination angles and polarization orientations. Looking from top to bottom shows the camera inclination angle with respect to the horizon approximately at 20°, 40°, and 60°. Looking from left to right, it starts with images captured without the polarizer, followed by orientation of the polarizer at approximately 0°, 45°, 90°, 135°. Legend: WPo denotes "without polarizer," Po denotes "with polarizer," θ_{Po} denotes "orientation of polarizer," and θ_{IC} denotes "inclination angle of camera looking down from the horizon."

pixel classification step. Denote $I_{i,j}^{t-\xi}, I_{i,j}^{t-\xi+1}, \ldots, I_{i,j}^{t}$ as the intensity values of a pixel at spatial location (i, j) for $\xi + 1$ consecutive frames. The first step involves the computation of a defined measure, termed *fluctuation measure*, with the formulation as follows:

$$\varpi_{i,j}^{t} = \frac{1}{SR} \sum_{u=0}^{\xi} \Xi_{i,j}^{u}, \quad \Xi_{i,j}^{u} = \begin{cases} 1, & \text{if } \delta_{i,j}^{t,u} > \Gamma \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\delta_{i,j}^{t,u}$ is the frame difference of the pixel between frame-t and frame-u, i.e., $\delta_{i,j}^{t,u} = \left| I_{i,j}^{t} - I_{i,j}^{u} \right|$, and SR is the sampling rate of video. Condition $\delta_{i,j}^{t,u} > \Gamma$ provides the measure of any significant change in intensity. Note that this formulation has shared some effective properties of the linear prediction model given in [39].

2) Filtering Scheme: With the computation of $\varpi_{i,j}^{t}$, the output of the classification and filtering steps are derived according to the following three conditional operations:

$$\boldsymbol{z}_{i,j}^{t} = \begin{cases} \boldsymbol{\psi}_{i,j}^{t}, & \varpi_{i,j}^{t} > \gamma_2 \\ \boldsymbol{\zeta}_{i,j}^{t}, & \gamma_1 < \varpi_{i,j}^{t} \leq \gamma_2 \\ \boldsymbol{x}_{i,j}^{t}, & \varpi_{i,j}^{t} \leq \gamma_1. \end{cases} \tag{13}$$

In the process, γ_1 and γ_2 are set to be one third and two thirds of ξ/SR, which has produced reasonably good results tested over long operational hours. It is noted that pixels with $\varpi_{i,j}^{t} > \gamma_2$ are those of fast moving parts of foreground or sensory noise. Therefore, we perform a mean filtering with a window of 3×3 on the respective color dimension for denoising purposes, giving the output $\boldsymbol{\psi}_{i,j}^{t}$. On the other hand, there is no filtering if $\varpi_{i,j}^{t} \leq \gamma_1$, which has avoided unnecessary blurring on the background. Therefore, $\boldsymbol{z}_{i,j}^{t}$ is the original color vector of the pixel, i.e., $\boldsymbol{z}_{i,j}^{t} = \boldsymbol{x}_{i,j}^{t}$. For pixels in the class of $\gamma_1 < \varpi_{i,j}^{t} \leq \gamma_2$, which contains foreground pixels and some reflective pixels, we aim to generate a "pseudocolor" based on spatiotemporal filtering operation for color compensation. Therefore, a set of mean filtered pixels W are collected from a number of successive previous frames at the same position, which contain no background pixels and reflections. The compensated color $\boldsymbol{\zeta}_{i,j}^{t}$ is generated by applying mean filtering on W. Such an operation has provided the estimation of color based on the average color of a pixel across consecutive frames, except the white color of water reflection.

V. Experimental Results

A real-time monitoring system has been set up at an outdoor olympic-sized swimming pool. Our system consists of a network of overhead cameras mounted surrounding the pool side with the combined view of all cameras providing a full coverage of the entire pool. Off-the-shelf CCTV cameras with auto-gain control are used and captured analogue videos are digitized live using frame grabbers. Running at near real time of 4 f/s from early morning to nighttime, the proposed algorithm operates under numerous outdoor situations amid glare, rain, and nighttime reflection.

A comprehensive study of the proposed system has been performed, comprising four aspects. Results of numerous case studies carried out are respectively reported in Sections V-A–V-D with outlines summarized as follows.

- Section V-A describes a case study when operating at whole day videos collected from early morning to nighttime. This evaluates the robustness of the system for long-hour operation. Qualitative evaluation is done by visually inspecting segmentation results of frames. Quantitative evaluation is computed in terms of true positive, false negative, and false positive on three selected segments of the whole day sequence. Each video segment depicts different scenarios at different time intervals.
- Section V-B reports a case study when operating at numerous challenging situations. In this section, five difficult situations that are frequently observed are highlighted, and insights into the task of swimmer detection are discussed.

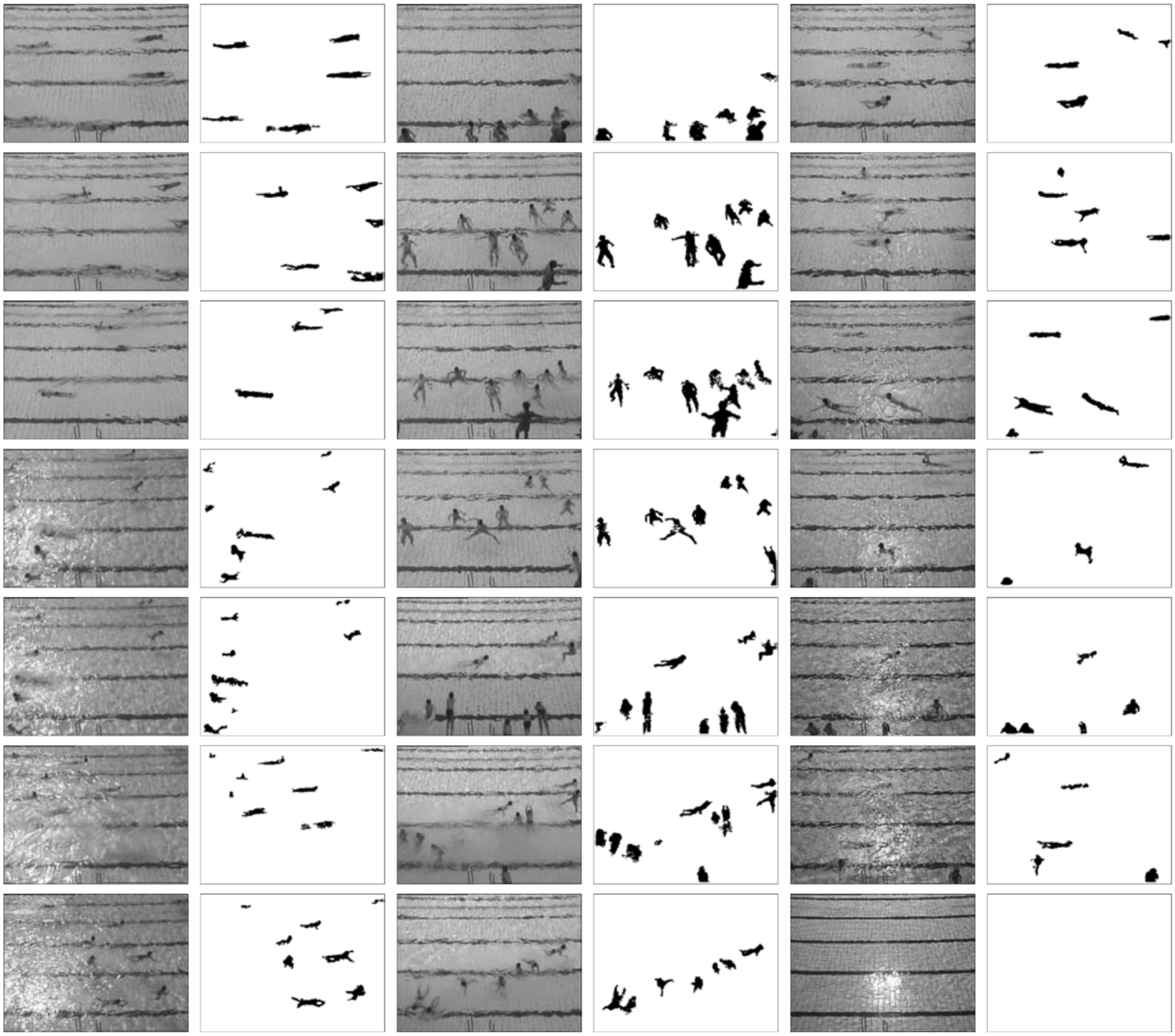


Fig. 9. (Top to bottom, left to right) The segmentations of swimmers on sample frames captured from numerous challenging scenarios at different time intervals from 8 am to 9 pm. Odd columns: Samples frames captured. Even columns: Corresponding segmentation results.

- Moving one step further, in Section V-C, we compare the performance of the proposed algorithm against state-of-the-art algorithms: the W4 method by Haritaoglu *et al.* [14] and the MoG method by Stauffer *et al.* [11], [12].
- In Section V-D, the proposed algorithm is tested on numerous indoor and outdoor video sequences for detecting objects-of-interest on the ground. This evaluates the generality of the proposed algorithm when deploying on environments beyond swimming pool.

A. Robustness for Continual Detection on a Whole Day Video

Fig. 9 provides a glimpse of various possibly situations involved when operating from 8 am to 9 pm on one typical day. Sample images (arranged from top to bottom, left to right) depict numerous difficult situations, such as sunlight reflection in the morning, high swimming activity during a swimming lesson, strong specular reflection at nighttime, and rapid change in ambient brightness occurs throughout the day. These frames show that background appears noisy and swimmers are occasionally almost invisible.

The even columns of Fig. 9 present results of segmentation for visual evaluation. It is noted that consistent performance have been observed over such an extended operational period. Images of the first two columns illustrate successful examples of detecting swimmers within reflective regions and showing immediate detection of these swimmers once they become visible. The third and fourth columns show results yielded for a crowded situation. Consistent detection performance is also observed when processing nighttime scenes as shown in the fifth and sixth columns, despite strong specular reflection.

In addition to qualitative evaluation, Fig. 10 presents the quantitative evaluation in terms of true positive, false negative, and false positive on three video sequences at different time intervals with length of 2000 frames each. The ground truth is obtained through visual inspection. We have manually gone through a checking of 6000 frames to derive the reported detection rates.

1) Fig. 10(a) presents detection results of a typical less-crowded scenario, which records a total of 1500 swimmers. In this case study, 1473 swimmers have been

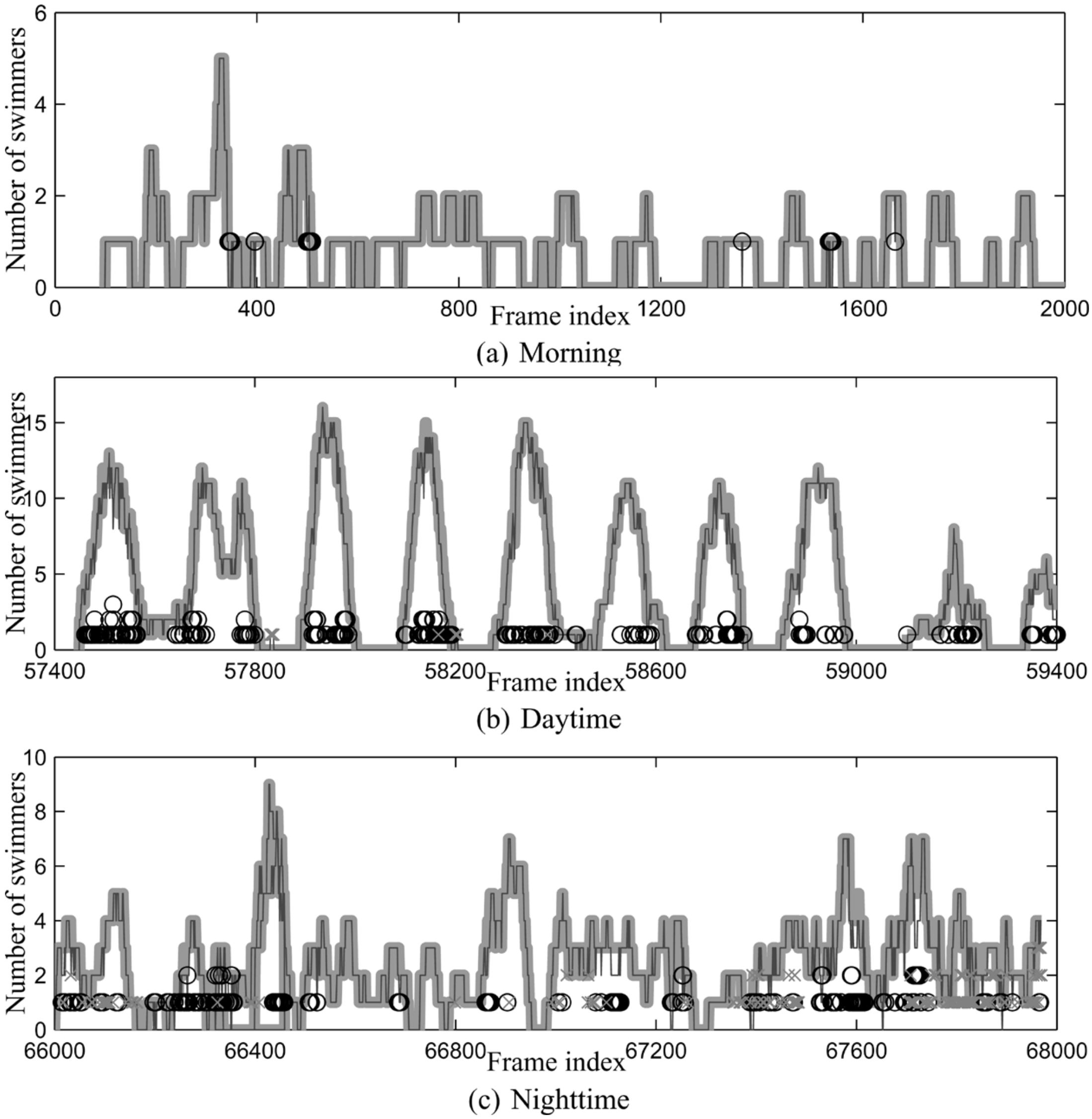


Fig. 10. Performance of the proposed algorithm evaluated on three selected video streams with lengths of 2000 frames each. They are captured at different time intervals within the same day. Results demonstrate a close correspondence between the number of detected swimmers obtained using the proposed algorithm and the results of the ground truth. Legend: Lighter line shows the ground truth of the number of swimmers in a frame; darker line shows the number of swimmers being detected; "o" indicates false negative; "×" indicates false positive.

correctly detected, achieving a detection rate of $\approx 98\%$ and showing that the detection curve follows closely the ground truth. The $\approx 2\%$ misdetection is mainly due to low contrast between swimmers and background, and also the small size of those swimmers, which has made the detection a difficult task. While maintaining a promising detection rate, the proposed algorithm is also robust in adapting to dynamic background movements of the background, which has achieved a $\approx 0\%$ false positive rate.

2) In contrast to the scenario in Fig. 10(a), Fig. 10(b) presents a more complicated scenario captured during a swimming lesson. It involves a detection process with a more vibrant background due to greater water disturbances. In this case study consisting of 8974 swimmers, the proposed algorithm records a $\approx 4\%$ false negative rate and a $\approx 0\%$ false positive rate, despite the higher complexity involved.

3) Furthermore, the algorithm also shows consistent results when operating at nighttime, regardless of strong specular reflection, as shown in Fig. 10(c). It yields a $\approx 4\%$ false negative and a $\approx 6\%$ false positive for such nighttime sequences that contains 5359 swimmers. The higher false positive compared with that of the daytime sequence is mainly due to a few cases of wrongly detecting background as foreground in the reflective region.

The respective results of true positive, false negative and false positive of the above case studies are also summarized in Table I.[1] In general, consistent performance is observed quantitatively across the different scenarios with a close correspondence between ground truth and detection results.

[1]The ground truth is obtained through visual inspection. We have manually gone through and checked 6000 frames to derive the reported detection rates.

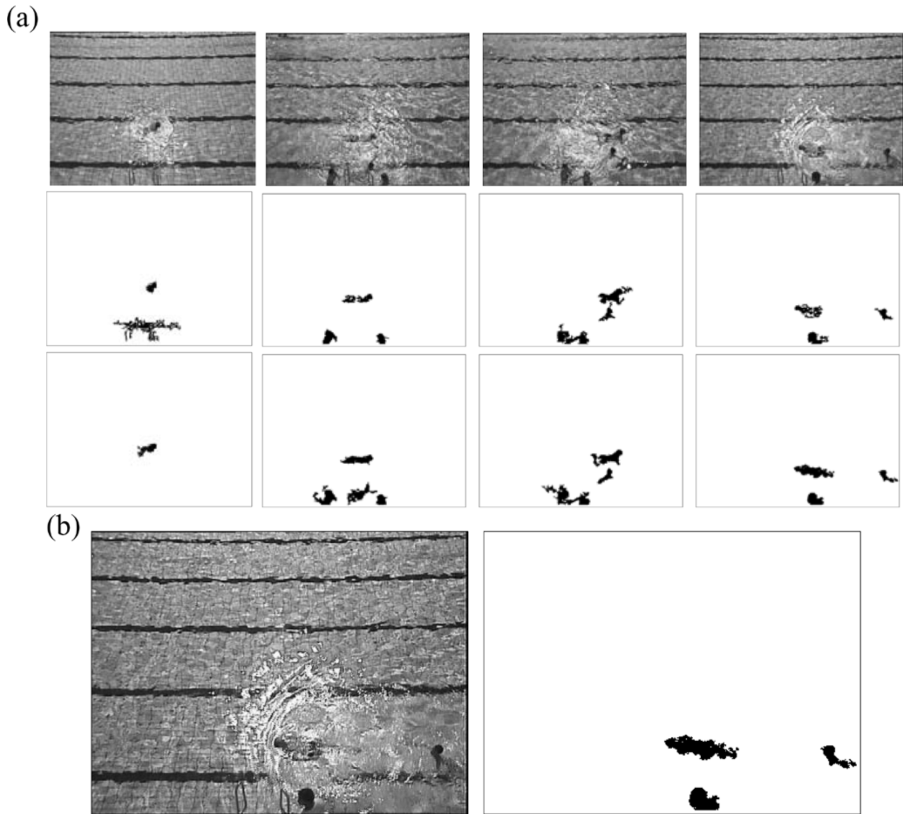


Fig. 11. (a) Comparison of detection results without and with filtering. First row: Sample frames of a typical nighttime sequence. Second row: Detection without filtering. Third row: Detection with the proposed spatiotemporal filtering scheme. (b) The enlargement of one image and its segmentation in (a) demonstrates the complexity involved under such badly illuminated condition.

TABLE I
QUANTITATIVE EVALUATION OF THE PROPOSED ALGORITHM IN TERMS OF TRUE POSITIVE, FALSE NEGATIVE, AND FALSE POSITIVE RATES WHEN EVALUATED ON THREE SELECTED VIDEO STREAMS OF FIG. 10. LEGEND: TP REPRESENTS TRUE POSITIVE; FN REPRESENTS FALSE NEGATIVE; FP REPRESENTS FALSE POSITIVE

	TP (%)	FN (%)	FP (%)
Morning	≈ 98	≈ 2	≈ 0
Daytime	≈ 96	≈ 4	≈ 0
Nighttime	≈ 96	≈ 4	≈ 6

B. *Reliability for Swimmers Detection on Challenging Scenarios*

In this section, we highlight five challenging scenarios that are faced. They include: 1) nighttime scenes with strong specular reflection; 2) rapid change in ambient lighting due to cloud movement; 3) rainy scenes with blurred foreground and background; 4) poorly visible swimmers when they are swimming along lane dividers; and 5) many swimmers playing in water that makes the background substantially disturbed. The first three situations relate to variations in ambient lighting, while the last two pertain to swimming activities in the pool. All these scenarios demand good performance in the aspects of eliminating ambient noise, fast adaption to ambient variation, and a high sensitivity of detection. We have evaluated the developed algorithm across all of these difficult scenarios.

Case 1: Nighttime Sequence With Specular Reflection: Specular reflection due to artificial lighting at nighttime poses the challenge that swimmers within reflective regions will experience deterioration in the visibility. Fig. 11 presents numerous sample images of a typical nighttime scene. As evident from the second and third rows of Fig. 11, improved performance with greater noise immunity and higher detection accuracy has been achieved after incorporating the proposed filtering scheme. It is noted that such a filtering step is also useful for removing reflection at daytime that exhibits the same momentary characteristic. However, as the condition of reflection at daytime is less severe and happens occasionally, such a step, is, thus, not activated during daytime for practical implementation.

Case 2: Sequence With Rapidly Changing Ambient Lighting: This problem demands a fast background updating mechanism to capture the background variations. At the same time, the system has to maintain the same sensitivity of foreground detection, but not blend them as background. Fig. 12 depicts an example of such a scenario, demonstrating a rapid change in brightness due to cloud movement. The figure depicts a substantial drop in the average intensity of a bright region bounded by a white box in between frames 1660 and frame 1770. It is equivalent to a period of 10 s. The failure to capture such variations in the background will result in unacceptable segmentation with the majority of these bright regions being misclassified as foreground. This problem has been accounted for in the framework by the proposed background updating mechanism.

Case 3: Rainy Day Sequence: The rainy day sequence, as depicted in Fig. 13, shows another unique environment. As seen

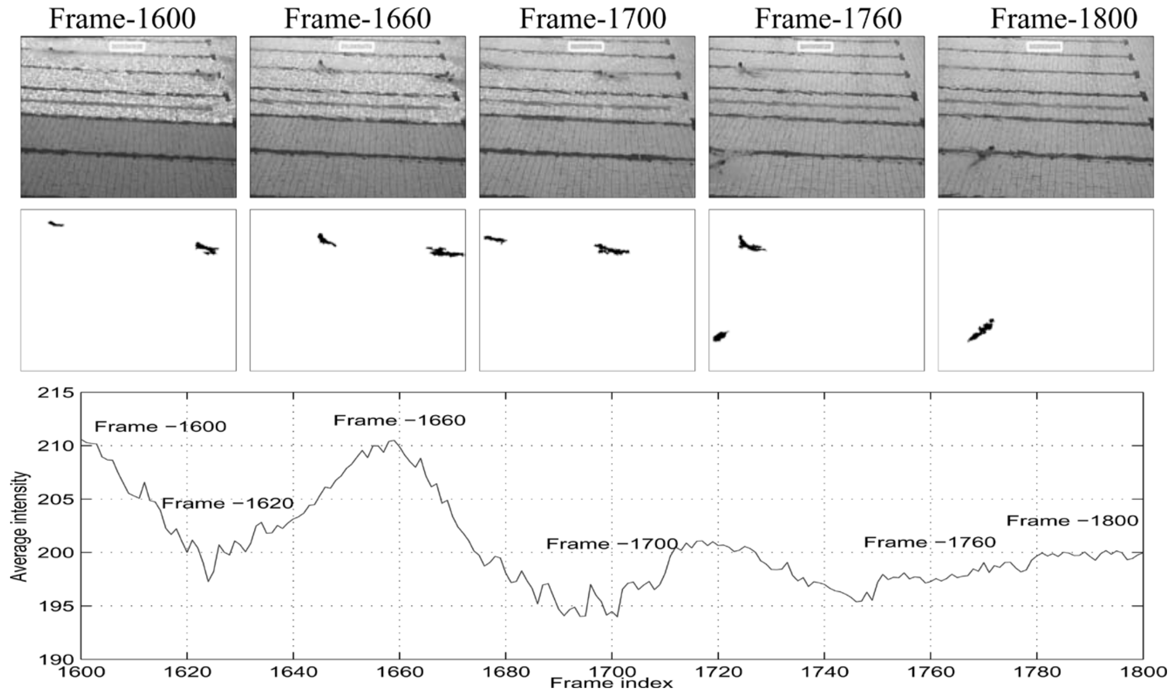


Fig. 12. Results of applying to a video stream which demonstrates a rapid change in the ambient brightness due to cloud movement. There are substantial drops in the brightness of a bright region bounded with white box between frame -1600 to frame -1620 and frame -1660 to frame -1700 (i.e., periods of 5 and 10 s, respectively, with a frame capturing rate of 4 f/s). This shows a need to have a fast background updating mechanism to avoid wrongly interpreting such background changes as foreground. First row: Sample frames captured. Second row: Detected swimmers. Third row: Plot of average intensity variation of the white bounding box over time.

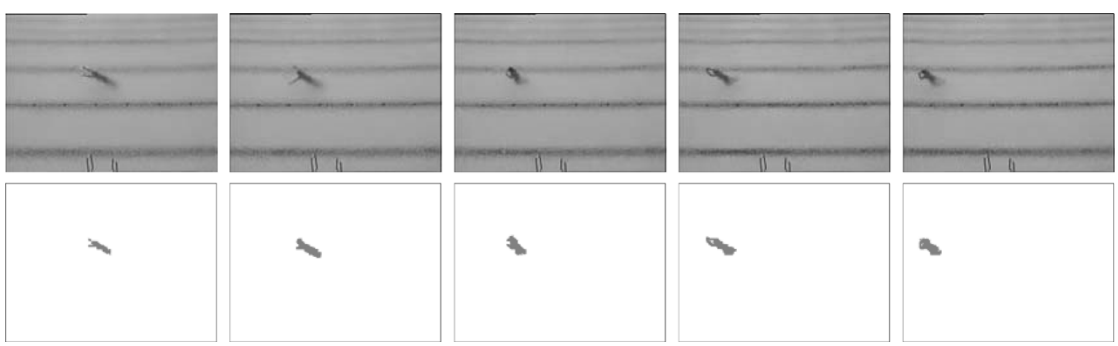


Fig. 13. Results of applying to a video stream of a rainy day. Rain has made captured images less clear, which, however, does not pose a major challenge to the algorithm.

from the figure, images captured during a rainy day are, in general, less sharp, compared with images obtained under a bright ambient lighting condition. The task of swimmer detection has technically become easier due to calmer (more stationary) backgrounds observed. In addition, from the operational aspect, accurate foreground detection is less critical during rainy periods, as swimming is generally not permitted.

Case 4: Close Color Similarity to Background: This problem arises due to the close color similarity between that of swimmers and black strips in between two swimming lanes, termed lane dividers, as shown in Fig. 14. The challenge of this problem is to find a balance between the need to have a highly sensitive thresholding process to detect small color discrepancy and the corresponding tradeoff that will lead to wrongly detecting portions of background becoming foreground noises. This has been accounted for in the framework by the proposed thresholding scheme.

The figure presents detection results when running the algorithm in two scenarios: 1) the swimmer continuously treading at the lane divider and 2) the swimmer swimming along the lane divider with poor visibility. As evident from Fig. 14, our results show a good performance yielded in terms of maintaining a highly sensitive detection while maintaining a low false error rate.

Case 5: Sequence With Swimmers Playing, Generating Great Background Disturbances: Fig. 15 depicts a scenario of having severe water disturbances. As shown in the figure, lane dividers become invisible after swimmers jump into the pool. Such a movement of water elements will be likely misinterpreted as foreground, if applying conventional approaches. In contrast, only foreground is detected, as shown in the results. The devised spatial matching mechanism has played the role of efficiently capturing such background movements. Furthermore, promising results have also been observed after the lane dividers

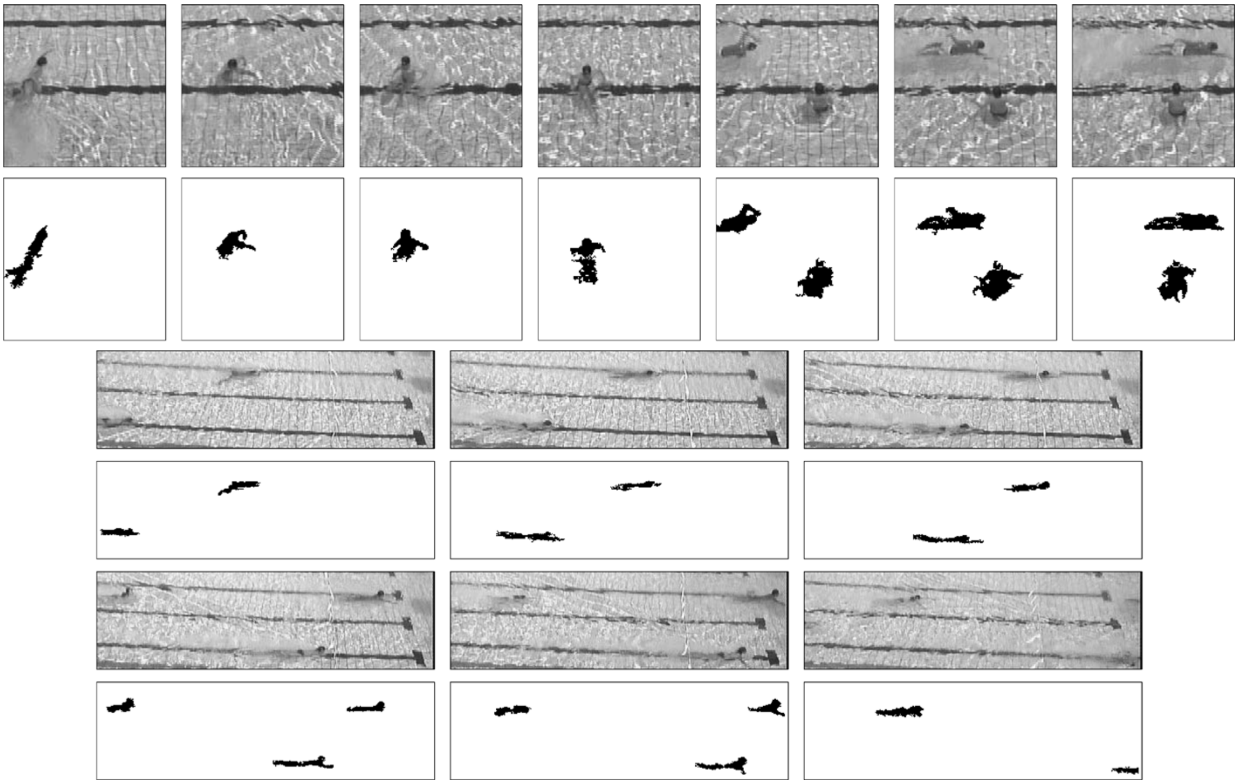


Fig. 14. Results of applying to sequences with swimmers treading on lane dividers (for the first two rows) and swimming along lane dividers (for the next four rows). Two major challenges faced are the weak visibility of swimmers that demand a highly sensitive thresholding and the corresponding tradeoff that will lead to wrongly detecting the lane divider as foreground.

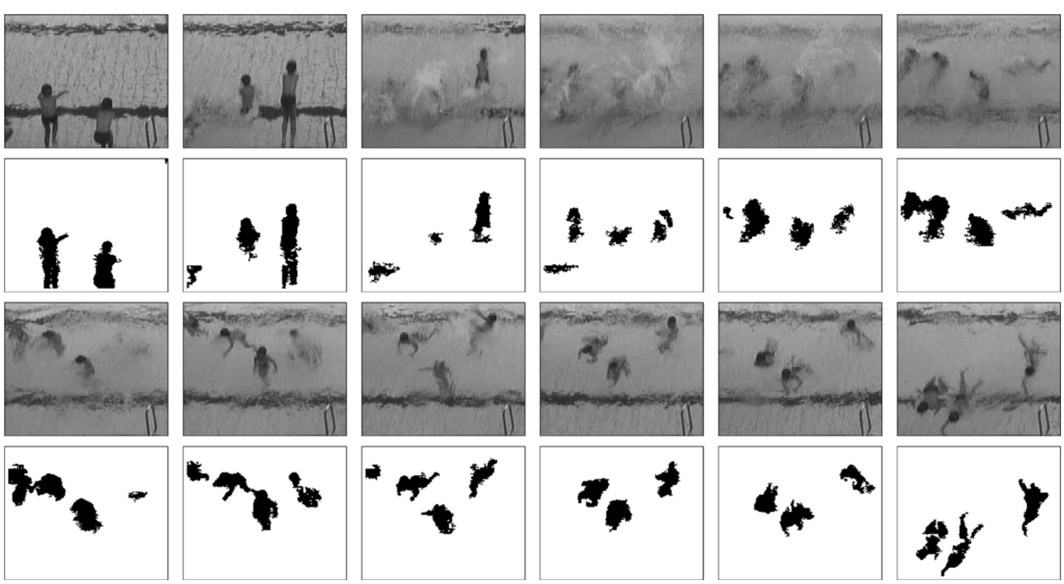


Fig. 15. Results of applying to video stream contains swimmers playing in swimming pools. Such high-movement activities have generated great disturbances on water surfaces and substantial changes to the background properties.

reappear again in the third and fourth rows, demonstrating the capability of correctly recognizing such a reappearance of lane dividers as background, but not foreground.

C. Performance Comparison

We also benchmarked the proposed algorithm against the well known W4 system [14] and the mixture of Gaussian (MoG) method by Stauffer *et al.* [11], [12]. When implementing the W4 and MoG methods, we have carefully fine tuned the parameter settings used in these methods to obtain the best results. The higher efficiency of our algorithm in detecting small and poorly visible swimmers, as well as maintaining a low false positive error (i.e., mis-classification of background elements to be foreground), has yielded better performance, when compared with the W4 and MoG methods, as shown in Fig. 16. The results demonstrate several successful examples of the system being

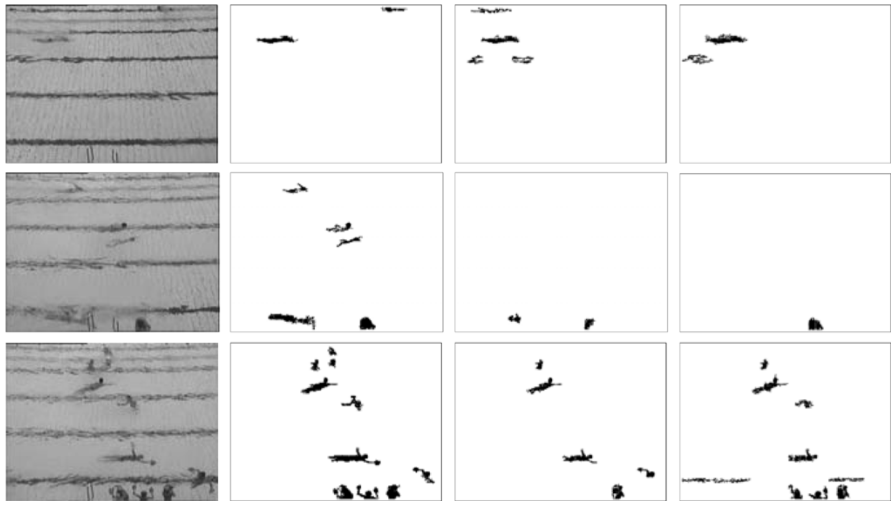


Fig. 16. Comparison of segmentation on a typical scene captured. The first column shows samples of scene captured. The second, third, and fourth columns present the corresponding segmentation results obtained by using our proposed algorithm W4 [14] and MoG method [11], [12], respectively.

able to detect small swimmers at various positions on the frame while maintaining a low false positive error, in which both W4 and MoG have attained less satisfactory performance. It is understandable that this may not be a fair comparison. We would like to highlight here that the current state-of-the-art algorithms may not be straightforwardly extended to such a new problem, and, naturally, there is a need for further research.

Table II presents the evaluation in terms of true positive, false positive, and false negative for two different scenarios: a typical less-crowded scenario in the first sequence and a crowded scenario during a swimming lesson in the second sequence. Compared with results of W4 and MoG methods, consistently better performance has been yielded by our algorithm. In addition, there is only slight deterioration in the performance for the second sequence.

TABLE II
COMPARISON OF DETECTION PERFORMANCES.
LEGEND: TP REPRESENTS TRUE POSITIVE;
FN REPRESENTS FALSE NEGATIVE; AND FP REPRESENTS FALSE POSITIVE

	Seq. #1			Seq. #2		
	TP(%)	FN(%)	FP(%)	TP(%)	FN(%)	FP(%)
Our algorithm	≈ 99	≈ 0	≈ 1	≈ 97	≈ 3	≈ 0
W4 [14]	≈ 90	≈ 10	≈ 9	≈ 62	≈ 38	≈ 0
MoG [11],[12]	≈ 97	≈ 3	≈ 2	≈ 73	≈ 26	≈ 6

D. Detection on the Ground at Outdoor Environments

Detecting the object-of-interest on the ground has been considered in most of the literature of automated surveillance. The typically considered environments encompass the following: on the road, offices, campuses, shopping malls, subway stations, etc. To show the generality of the proposed algorithm for detecting foreground on the ground, rather than limited to swimming pool environments, we have tested the algorithm on three selected video streams with selected results presented in Fig. 17.

Generally, the proposed algorithm shows promising results as most of the foreground objects have been accurately detected with reasonably good results. This, thus, demonstrates its generality to be exploited to other environments rather than confined to swimming pool environments.

VI. CONCLUSION

In this paper, the problem of automated monitoring based on video surveillance approaches in noisy and highly dynamic aquatic environments has been discussed. In particular, this paper has addressed, in great detail, two fundamental issues: 1) the issue of devising a good modeling scheme for a dynamic aquatic background, and 2) the issue of recovering camouflaged foregrounds for frame enhancement prior to background subtraction.

The first issue has been jointly addressed in this paper using a proposed region-based background modeling and a devised thresholding scheme. In the proposed method, instead of modeling the background pixelwise or blockwise, as conventionally suggested, we have proposed the modeling of background as a composition of homogeneous region processes. This models the dynamic aquatic background that exhibits random homogeneous blob movements due to continual disturbance at the water's surface and experiences illumination and color variations due to changes in outdoor ambient brightness over time. In the devised thresholding scheme, we have combined the principle of thresholding with hysteresis, denoising, and connected component grouping under an integrated framework. This proposes a better alternative to the intuitive single thresholding approach, addressing the objective to attain a good tradeoff between maximizing target detection while minimizing background noises. Meanwhile, the second issue has been addressed by a proposed preprocessing filtering scheme based on a defined fluctuation measure. It is defined to classify each pixel of a current frame into different pixel types, such that appropriate filtering actions, i.e., color compensation, could be applied accordingly. Such a filtering scheme has been incorporated as a preprocessing step for frame enhancement during nighttime operation.

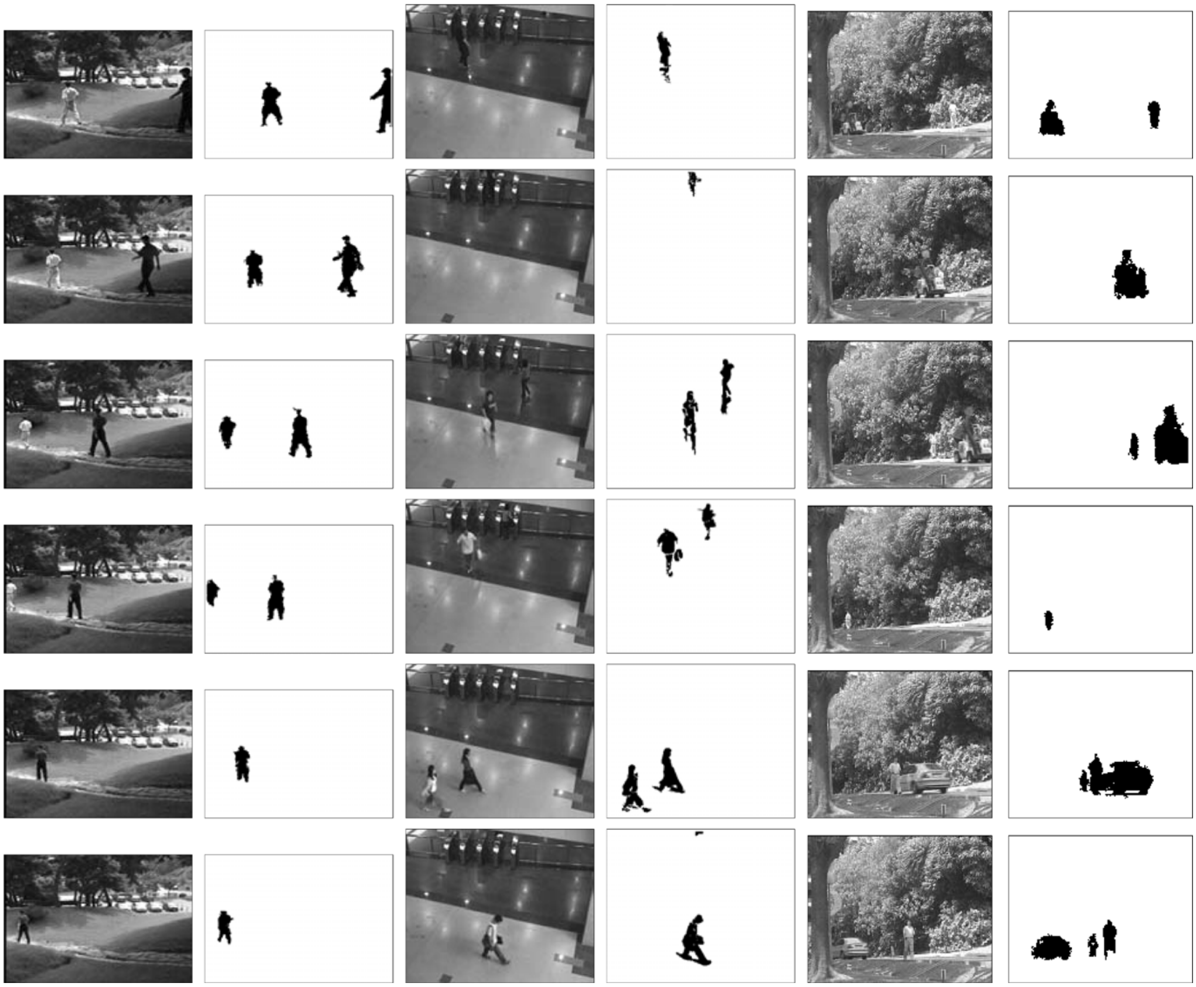


Fig. 17. (Top to bottom, left to right) Results of applying the proposed algorithm on various indoor and outdoor video sequences for detecting foreground objects on the ground. The promising results shown above demonstrate the generality of the proposed algorithm to be exploited for other applications beyond the swimming pool environment.

The experimental results show that the developed framework is both robust under long operation hours and reliable in handling numerous challenging situations. In addition, as evident from the results, when detecting objects on the ground, such developed framework can also be considered for the use on other commonly encountered surveillance problems.

One great potential application of the discussed research pertains to providing constant monitoring of swimmers at a swimming pools that could improve the safety of our surrounding aquatic facilities. Any occurrence of drowning incidents could then be alerted in the first instance. In order to develop a robust and practical system for early drowning symptom detection at an outdoor public swimming pool, we propose further investigation on the following topics:

- an advanced event-inference module for complex swimming behavioral analysis;
- a crowd density estimation mechanism that estimates the density of a crowd in different zones covered by different cameras (this is useful to lifeguards in focusing on higher risk regions);
- drowning detection for crowded scenarios, where occlusion is an issue of concern.

Acknowledgment

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